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Experiment Title : Demonstration of Router Hardware and Connecting Different LANs

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Experiment No. 07: Demonstration of Router Hardware and Connecting Different LANs

Aim

To identify the hardware interfaces of a Router and to design, configure, and verify the connectivity between two distinct Local Area Networks (LANs) using a Router in Cisco Packet Tracer.

Theory

1) What is a Network Router?

A **network router** is a networking device that connects **multiple networks** together and forwards data between them.

It operates at **Layer 3 (Network Layer)** of the OSI model.

It uses **IP addresses** to determine where data should go.

It connects **local networks (LANs)** to other networks, such as the Internet.

Example:

When you open a website, your home router sends your request from your local network to your Internet Service Provider (ISP).

2) Role of a "Default Gateway" in a Network

A **default gateway** is the device (usually a router) that a computer uses to send data **outside its local network**.

Why it is needed:

A computer can directly communicate with devices in the same subnet.

But to reach devices in other networks (like the Internet), it sends the data to the **default gateway**.

The gateway then forwards the data to the correct external network.

Example:

If your PC's IP is 192.168.1.10 and it wants to reach 8.8.8.8: It

sends the packet to its **default gateway** (e.g., 192.168.1.1)

The router then forwards it toward the Internet.

Without a default gateway, devices cannot communicate outside their local network.

3) How Does a Router Differ from a Switch?

Feature	Router	Switch
OSI Layer	Layer 3 (Network)	Layer 2 (Data Link)

Uses	IP addresses	MAC addresses
Connects	Different networks	Devices within same network
Internet access	Yes	No (alone)
Routing decision	Based on routing table	Based on MAC address table

Simple Explanation:

Switch → Connects devices inside a LAN (PCs, printers).

Router → Connects different networks together (LAN to Internet).

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4) What is a Routing Table and How is it Used?

A **routing table** is a list stored inside a router that contains information about:

- Destination networks
- Next hop addresses
- Interface to use
- Metric (cost of route) How it Works:

When a router receives a packet:

It checks the **destination IP address**.

It compares it with entries in the routing table.

It selects the best matching route.

It forwards the packet to the next hop or final destination.

Example Routing Table Entry:

Destination	**Subnet Mask**	**Gateway**	**Interface**
192.168.1.0	255.255.255.0	Direct	LAN
0.0.0.0	0.0.0.0	192.168.1.1	WAN

0.0.0.0 represents the **default route** (used when no specific match is found).

Hardware Components Required

- **PC-PT (x6):** End devices acting as network nodes (3 for LAN 1, 3 for LAN 2).
- **Switch 2960 (x2):** Two 24-port switches to create independent local networks. One switch for each LAN.
- **Router ISR 4321 (x1):** The device to connect the two LANs.
- **Cabling:** Copper Straight-Through cables for all connections.

Hardware and Connection Setup

- LAN 1 (Network ID): 192.168.1.0 | Subnet Mask: 255.255.255.0
- LAN 2 (Network ID): 10.0.0.0 | Subnet Mask: 255.0.0.0
- PC Naming Convention:
  - LAN 1: S11_01_A, S11_01_B, S11_01_C (IPs: .2, .3, .4) ○

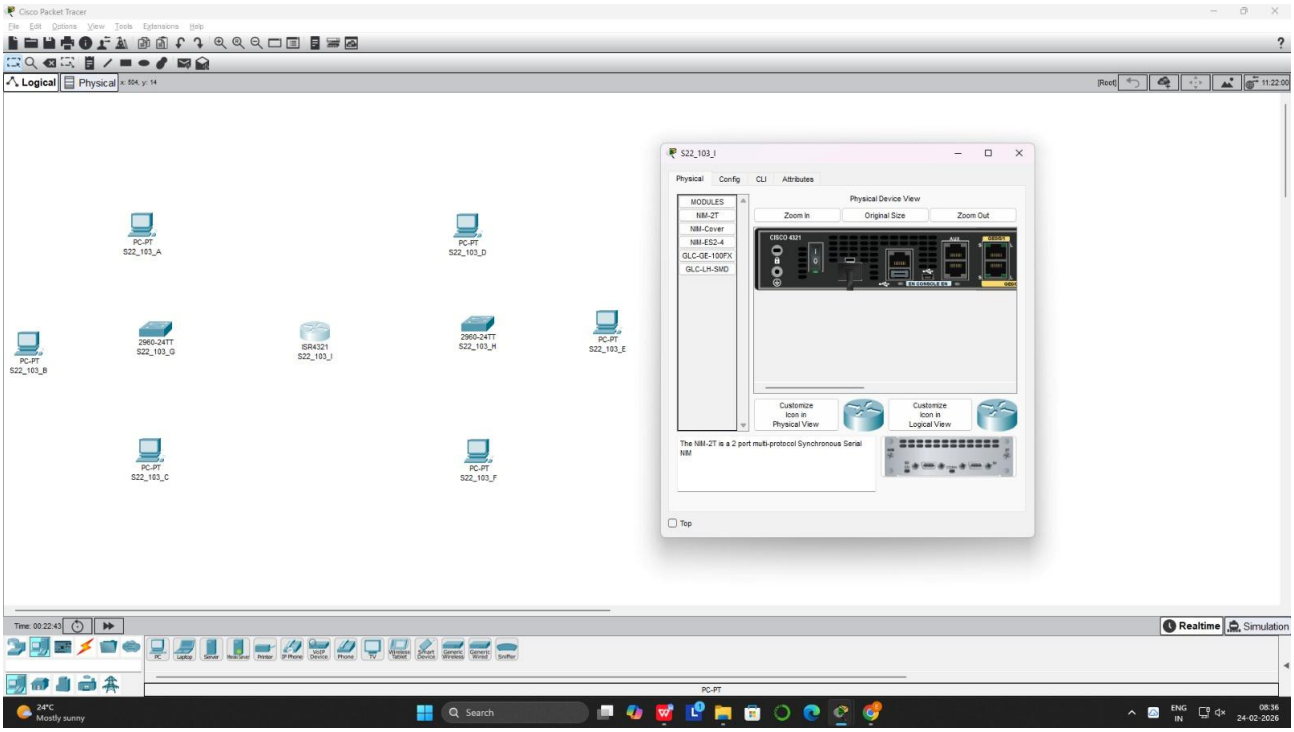
  LAN 2: S11_01_D, S11_01_E, S11_01_F (IPs: .2, .3, .4)

Procedure Step 1: Physical Topology Design

- Place the ISR 4321 Router in the center of the workspace.
- Place two 2960 Switches (one for each LAN) and six PCs (three per switch).
- Naming: Rename PCs following the convention (e.g., S11_01_A through S11_01_F).

Step 2: Hardware Identification

- Click on the Router and select the Physical tab.
- Observe the modular slots and the GigabitEthernet ports.
- Note the power switch; ensure the router is turned ON.



Step 3: Physical Cabling

- Connect Switch 0 to Router interface GigabitEthernet 0/0/0.

- Connect Switch 1 to Router interface GigabitEthernet 0/0/1.
- Connect all PCs to their respective switches.
- Observation: Note that the link lights between the switches and the router are Red. This is because router ports are disabled by default.

Step 4: Router Interface Configuration (The "Bridge" Setup)

1. Click **Router > CLI**. Type 'no' at the initial setup prompt.
2. Enter the following commands to configure the gateways:
   - Router> enable
   - Router# config t
   - Router(config)# interface gigabitEthernet 0/0/0
   - Router(config-if)# ip address 192.168.1.1 255.255.255.0
   - Router(config-if)# no shutdown
   - Router(config-if)# interface gigabitEthernet 0/0/1
   - Router(config-if)# ip address 10.0.0.1 255.0.0.0 Router(config-if)# no shutdown

Observation: The link lights should now turn Green.

Command	Purpose / Function
enable	Moves the user from **User EXEC Mode** (view only) to **Privileged EXEC Mode**. This allows the user to view detailed configurations and enter setup modes.
config t (configure terminal)	Moves the router into **Global Configuration Mode**. This is the only mode where you can make changes that affect the router as a whole.
interface gigabitEthernet 0/0/0	Moves the router into **Interface Configuration Mode**. It tells the router, "The next commands I type are only for this specific physical port."
ip address [IP] [Mask]	Assigns a logical Identity (IP) and defines the network size (Subnet Mask) for that interface. This becomes the **Default Gateway** for that LAN.
no shutdown	By default, Cisco router ports are "Administratively Down" (turned off for security). This command **powers on** the interface to allow data flow.

Step 5: IP Addressing (Logical Setup)

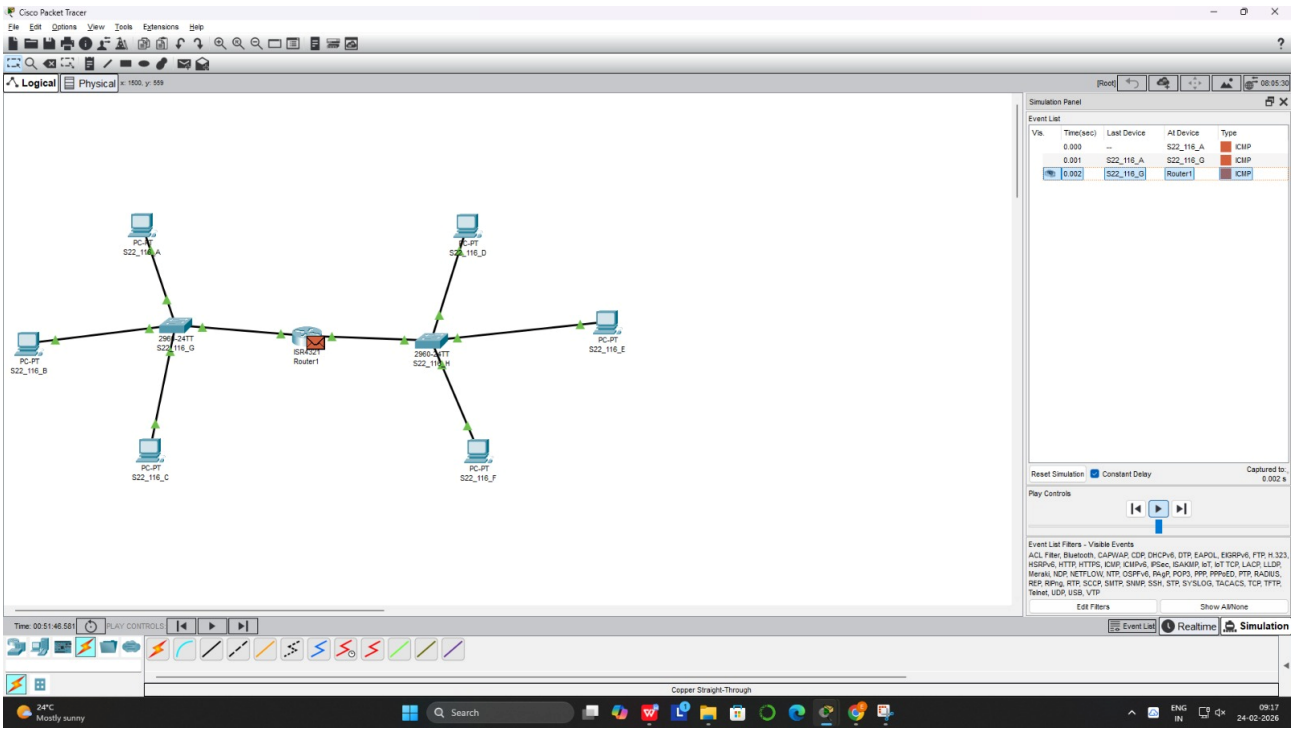
- **For LAN 1 PCs:** Set IPs 192.168.1.x and set Default Gateway to 192.168.1.1.
- **For LAN 2 PCs:** Set IPs 10.0.0.x and set Default Gateway to 10.0.0.1.

Step 6: Testing and Simulation

1. **Real-time:**

- Open Command Prompt on **S11_01_A** and type ping 10.0.0.4.
- Verify connectivity. (Note: The first ping may timeout due to ARP).

2. **Simulation Mode:**
   - Filter for ICMP. Use the Add Simple PDU (Envelope) to click S11_01_A (LAN 1) then S11_01_F (LAN 2).
   - Observe the packet crossing from the 192.x network to the 10.x network through the router.



Observations

- **Gateway Necessity:** If the "Default Gateway" is not configured on the PCs, the packet never reaches the router and communication fails.
- **Routing Decision:** The Router stops the broadcast from LAN 1 and intelligently forwards the packet specifically to the interface connected to LAN 2.
- **Different Subnets:** Communication was successfully established between two completely different IPs, which is impossible with only a switch.

Conclusion

The experiment successfully demonstrates that a **Router** is the essential hardware for Inter-LAN or Inter-Network communication. It proves that while switches manage local traffic, routers are required to cross network boundaries by using logical IP addresses.